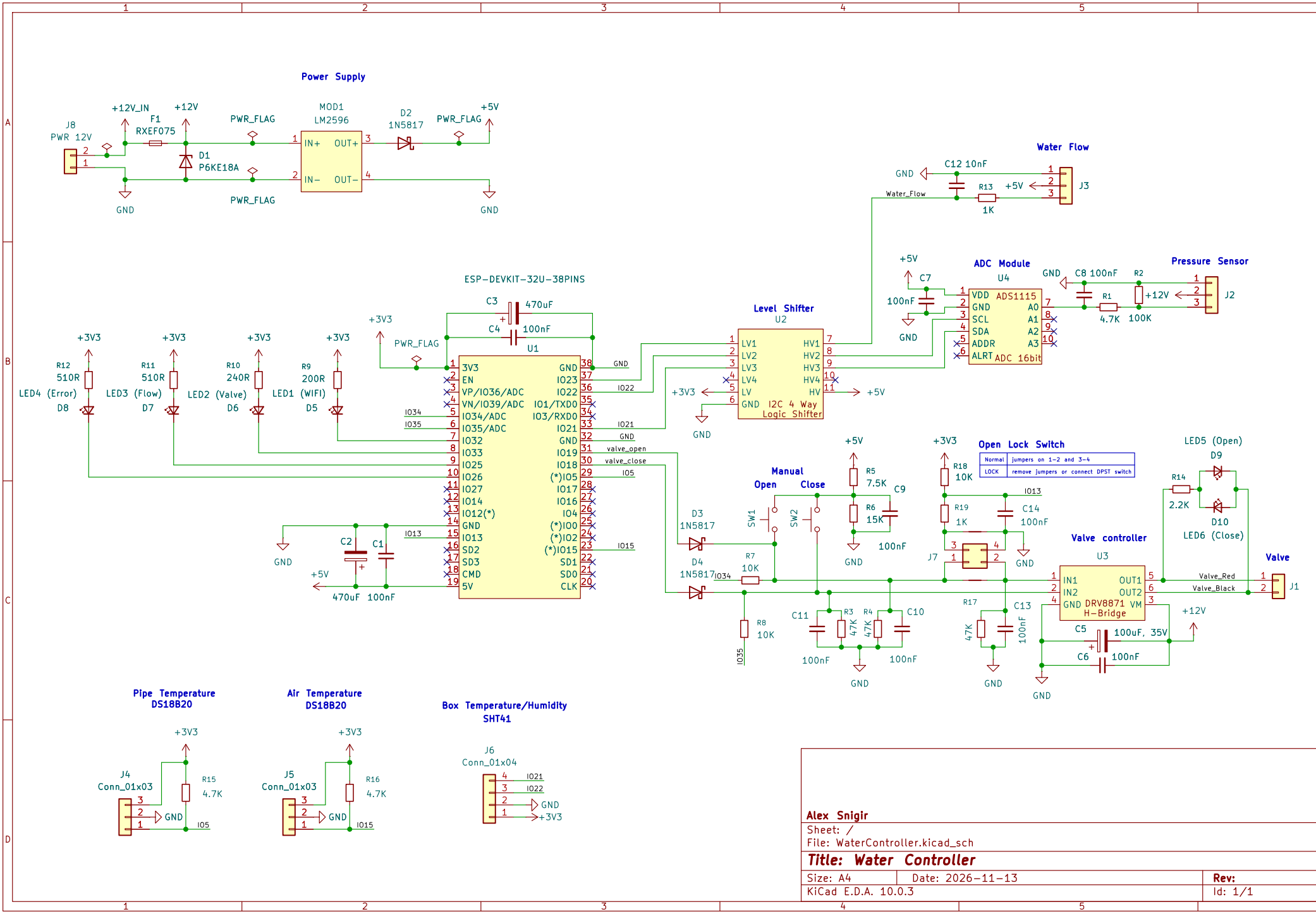

Power Supply
J8
PWR 12V
+12V_IN
F1
RXEF075
+12V
D1
P6KE18A
PWR_FLAG
MOD1
LM2596
IN+
OUT+
IN−
OUT−
D2
1N5817
PWR_FLAG
+5V
GND
ESP−DEVKIT−32U−38PINS
U1
C3
470uF
C4
100nF
+3V3
PWR_FLAG
3V3
EN
VP/IO36/ADC
VN/IO39/ADC
IO34/ADC
IO35/ADC
IO32
IO33
IO25
IO26
IO27
IO14
IO12(*)
GND
IO13
SD2
SD3
CMD
5V
GND
IO23
IO22
IO1/TXD0
IO3/RXD0
IO21
GND
IO19
IO18
(*)IO5
IO17
IO16
IO4
(*)IO0
(*)IO2
(*)IO15
SD1
SD0
CLK
C2
C1
470uF
100nF
R12
510R
LED4 (Error)
D8
R11
510R
LED3 (Flow)
D7
R10
240R
LED2 (Valve)
D6
R9
200R
LED1 (WIFI)
D5
Level Shifter
U2
LV1
LV2
LV3
LV4
LV
GND
HV1
HV2
HV3
HV4
HV
I2C 4 Way
Logic Shifter
Water Flow
C12 10nF
R13
1K
J3
Water_Flow
ADC Module
U4
C7
100nF
VDD ADS1115
GND
SCL
SDA
ADDR
ALRT
A0
A1
A2
A3
ADC 16bit
C8 100nF
R1
4.7K
R2
100K
Pressure Sensor
J2
Manual
Open
Close
SW1
SW2
D3
1N5817
D4
1N5817
R7
10K
R8
10K
R5
7.5K
R6
15K
C9
100nF
C11
100nF
R3
47K
R4
47K
C10
100nF
Open Lock Switch
Normal
jumpers on 1−2 and 3−4
LOCK
remove jumpers or connect DPST switch
R18
10K
R19
1K
J7
C14
100nF
R17
47K
C13
100nF
Valve controller
U3
IN1
IN2
GND DRV8871 VM
H−Bridge
OUT1
OUT2
C5
100uF, 35V
C6
100nF
LED5 (Open)
D9
R14
2.2K
D10
LED6 (Close)
Valve
J1
Valve_Red
Valve_Black
Pipe Temperature
DS18B20
J4
Conn_01x03
R15
4.7K
Air Temperature
DS18B20
J5
Conn_01x03
R16
4.7K
Box Temperature/Humidity
SHT41
J6
Conn_01x04
Alex Snigir
Sheet: /
File: WaterController.kicad_sch
Title: Water Controller
Size: A4
Date: 2026−11−13
Rev:
KiCad E.D.A. 10.0.3
Id: 1/1